

E-Commerce Data Analysis Report

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Tools Used: SQL Server, SQL

1. Objective

The main aim of this project was to analyze e-commerce data using SQL Server and understand customer behaviour, sales performance, delivery efficiency, and seller contribution.

2. Dataset Description

The project used multiple tables, such as Customers, Orders, Payments, Products, Sellers, Reviews, and Product Categories. Relationships between the tables were created using primary keys and foreign keys to maintain proper database structure.

3. Methodology

Different SQL concepts were used to perform the analysis, including joins, aggregate functions, CTEs, subqueries, views, indexes, and window functions. The queries were written to identify trends and generate business insights from the dataset.

4. Key Findings

Customer Behavior

Observation: Most customers placed only one order. This shows that repeat purchases were comparatively low.

Suggestion: Customer retention strategies such as offers and loyalty programs can help increase repeat orders.

Revenue Analysis

Observation: The overall revenue increased along with the number of orders placed.

Suggestion: Tracking revenue regularly can help the business understand sales performance better.

Monthly Sales Trends

Observation: Order volume changed across different months, showing seasonal sales patterns.

Suggestion: Businesses can plan inventory and promotions based on peak sales periods.

Seller Performance

Observation: A small group of sellers generated a large share of the total revenue.

Suggestion: Supporting lower-performing sellers may help improve overall platform performance.

Product Category Analysis

Observation: Some product categories performed better in terms of sales and order volume.

Suggestion: High-performing categories can be prioritized for marketing and stock management.

Payment Method Analysis

Observation: Customers preferred only a few payment methods more frequently than others.

Suggestion: Improving popular payment options can provide a smoother customer experience.

Delivery Performance

Observation: Several orders were delivered after the estimated delivery date.

Suggestion: Improving delivery operations can reduce delays and improve customer satisfaction.

Customer Satisfaction

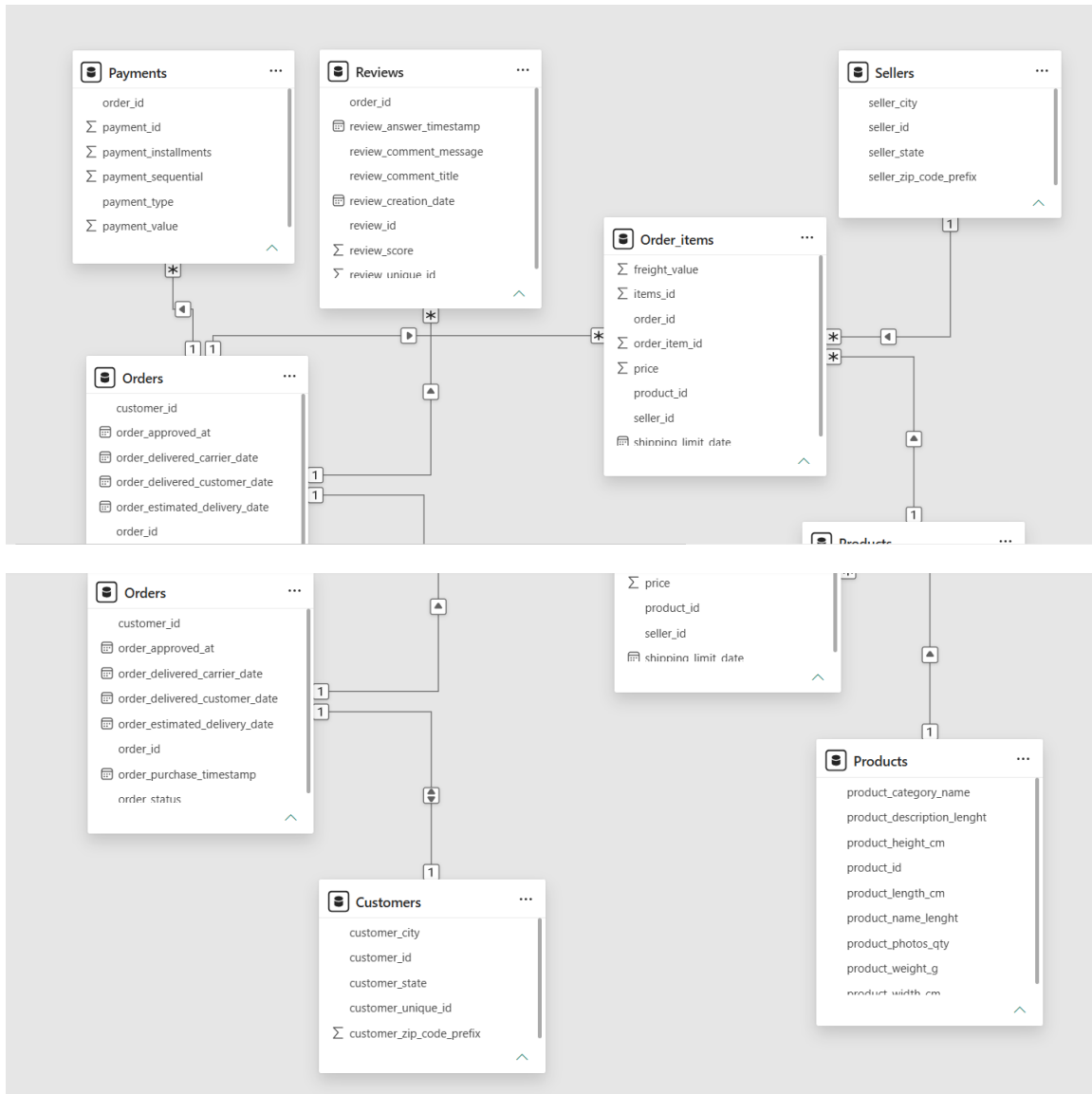
Observation: Review scores showed that delivery time and product quality affected customer experience.

Suggestion: Maintaining product quality and reducing delays can improve customer ratings.

5. SQL Techniques Used

- Joins to combine multiple tables
- Aggregate functions for revenue and trend analysis
- CTEs for organizing complex queries
- Subqueries for filtering data
- Window functions for ranking sellers
- Views for reusable reporting queries
- Indexes to improve query performance
- Primary and foreign keys for database relationships

6. DATABASE SCHEMA



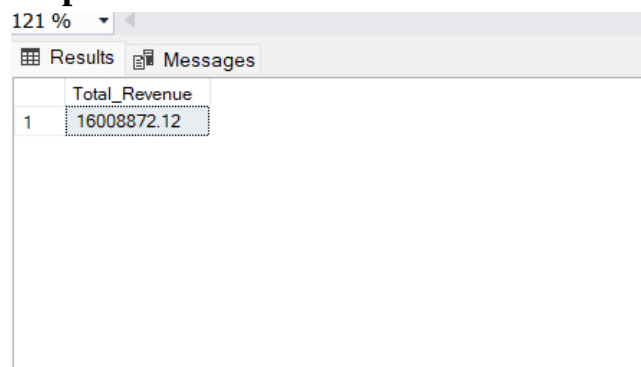
7. Key Business Analysis Queries¹.

-- Total Revenue generated by the platform

```
select round(sum(payment_value),2) AS Total_Revenue
```

```
FROM Payments;
```

output:-



The screenshot shows a database interface with a zoom level of 121%. It displays a table with one column, 'Total_Revenue', and one row containing the value '16008872.12'. The interface includes tabs for 'Results' and 'Messages'.

	Total_Revenue
1	16008872.12

2.

-- Top sellers by revenue

```
select * from Sellers ;
```

```
select * from Payments ;
```

```
select * from Order_items;
```

```
select top(10) s.seller_id,  
round(sum(p.payment_value),2) as revenue  
from Order_items o join Sellers s  
on o.seller_id = s.seller_id  
join Payments p  
on p.order_id = o.order_id  
group by s.seller_id  
order by round(sum(p.payment_value),2) desc
```

output:-

121 %

Results Messages		
	seller_id	revenue
1	7c67e1448b00f6e969d365cea6b010ab	507166.91
2	1025f0e2d44d7041d6cf58b6550e0bfa	308222.04
3	4a3ca9315b744ce9f8e9374361493884	301245.27
4	1f50f920176fa81dab994f9023523100	290253.42
5	53243585a1d6dc2643021fd1853d8905	284903.08
6	da8622b14eb17ae2831f4ac5b9dab84a	272219.32
7	4869f7a5dfa277a7dca6462dcf3b52b2	264166.12
8	955fee9216a65b617aa5c0531780ce60	236322.3
9	fa1c13f2614d7b5c4749cbc52fecda94	206513.23
10	7e93a43ef30c4f03f38b393420bc753a	185134.21

3.

-- Monthly sales trends

select

YEAR(order_purchase_timestamp) AS Order_Year,

MONTH(order_purchase_timestamp) AS Order_Month,

COUNT(order_id) AS Total_Orders

from Orders

group by YEAR(order_purchase_timestamp),

MONTH(order_purchase_timestamp)

order by Order_Year, Order_Month;

output:-

.21 % ▾

Results			
	Order_Year	Order_Month	Total_Orders
1	2016	9	4
2	2016	10	324
3	2016	12	1
4	2017	1	800
5	2017	2	1780
6	2017	3	2682
7	2017	4	2404
8	2017	5	3700
9	2017	6	3245
10	2017	7	4026
11	2017	8	4331
12	2017	9	4285
13	2017	10	4631
14	2017	11	7544
15	2017	12	5673
16	2018	1	7269
17	2018	2	6728
18	2018	3	7211
19	2018	4	6939
20	2018	5	6873
21	2018	6	6167
22	2018	7	6292
23	2018	8	6512
24	2018	9	16
25	2018	10	4

4.

-- customer retention analysis

select

customer_id,

COUNT(order_id) AS Total_Orders

from Orders

group by customer_id

having COUNT(order_id) > 1

order by Total_Orders DESC;

output:-



121 %

Results Messages

customer_id	Total_Orders
-------------	--------------

5.

--Rank sellers based on total revenue.

select * from Order_items

select seller_id,

sum(price + freight_value) as revenue,

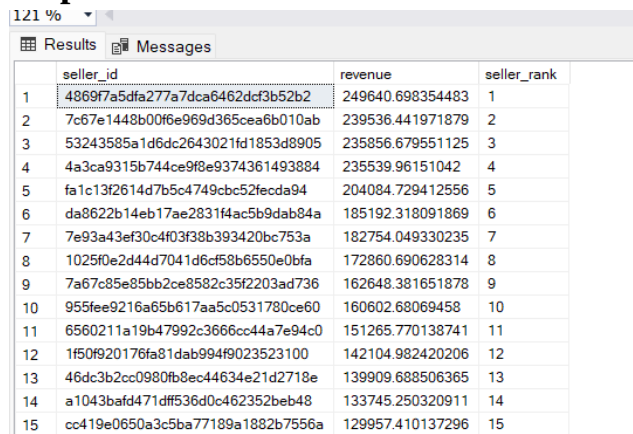
DENSE_RANK() over(order by sum(price + freight_value) desc) as seller_rank

from Order_items

group by seller_id;

go

output:-



121 %

Results Messages

	seller_id	revenue	seller_rank
1	4869f7a5dfa277a7dca6462dcf3b52b2	249640.698354483	1
2	7c67e1448b00f6e969d365cea6b010ab	239536.441971879	2
3	53243585a1d6dc2643021fd1853d8905	235856.679551125	3
4	4a3ca9315b744ce9f8e9374361493884	235539.96151042	4
5	fa1c13f2614d7b5c4749cbc52fecda94	204084.729412556	5
6	da8622b14eb17ae2831f4ac5b9dab84a	185192.318091869	6
7	7e93a43ef30c4f03f38b393420bc753a	182754.049330235	7
8	1025f0e2d44d7041d6cf58b6550e0bfa	172860.690628314	8
9	7a67c85e85bb2ce8582c35f2203ad736	162648.381651878	9
10	955fee9216a65b617aa5c0531780ce60	160602.68069458	10
11	6560211a19b47992c3666cc44a7e94c0	151265.770138741	11
12	1f50f920176fa81dab994f9023523100	142104.982420206	12
13	46dc3b2cc0980fb8ec44634e21d2718e	139909.688506365	13
14	a1043bafd471dff536d0c462352beb48	133745.250320911	14
15	cc419e0650a3c5ba77189a1882b7556a	129957.410137296	15

6.

-- Delayed deliveries or late orders

```
select * from Orders
```

```
select order_id,
```

```
order_delivered_customer_date,
```

```
order_estimated_delivery_date from Orders
```

```
where order_delivered_customer_date > order_estimated_delivery_date
```

```
SELECT COUNT(*) AS delayed_orders
```

```
FROM Orders
```

```
WHERE order_delivered_customer_date > order_estimated_delivery_date;
```

output:-

1.21 %

Results		Messages																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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8. Conclusion

This project helped in understanding how SQL can be used for real-world business analysis. The analysis provided useful insights into customer activity, sales trends, seller performance, delivery issues, and customer satisfaction.